Dong Hyeon Park

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EDUCATION

- **Yonsei University** Mar. 2025 –
  M.S. degree candidate, Mechanical Engineering Seoul, Republic of Korea
  Advisor: Prof. Jun Young Yoon
  GPA: 3.97/4.0

  Selected Coursework: Signals and Systems, Precision Manufacturing System, Optimal Control and Reinforcement Learning, Theory of Automatic Control, Nonlinear Control, Special Topics in Manufacturing Processes, Manufacturing System Design

- **Yonsei University** Mar. 2022 - Feb. 2025
  B.S., Mechanical Engineering Seoul, Republic of Korea
  GPA: 3.82/4.0, **(Rank: 3rd out of 150, Top 2%)**

  Thesis: Modeling Force–current Formula Including Coupling Term Between DQ-axis Current and Analysis of Force Ripple in Iron-core Permanent Magnet Linear Synchronous Motor *(Advisor: Prof. Jun Young Yoon)*

  Selected Coursework: Mechatronics, Mechanical System Control, Creative Product Design, Manufacturing Process, Independent Study, Design of Mechanical Element, Mechanism Design

- **Kyunghee University (Transferred to Yonsei University)** Mar. 2018 - Feb. 2022
  B.S., Mechanical Engineering (Military: Mar. 2019 – Oct. 2020) Seoul, Republic of Korea
  GPA: 3.74/4.0

RESEARCH INTERESTS

- Precision Mechatronics Systems
- Actuators
- Precision Motion Control
- Stiffness Tunable Mechanisms
- Additive Manufacturing

RESEARCH EXPERIENCES

Research of Manufacturing & Mechatronics Laboratory *(Advisor: Prof. Jun Young Yoon)* Mar. 2025 –
Yonsei University Seoul, Republic of Korea

1. "AlNiCo-based Active Stiffness Tunable XY Stage with Resonance-matched High-gain Control for Ultra-low power Large-area Scanning Motion" (Personal researcher)

- Proposed an AlNiCo-based active stiffness tunable stage enabling resonance matching for ultra-low power large-area scanning motion.
- Implemented stiffness tuning by AlNiCo-based tunable magnets.
- Achieved up to a **32.43X** reduction in energy consumption compared to non-stiffness tunable stage.

2. "Design and Control of Monolithic Hybrid Reluctance Actuator for Long-stroke In-plane 3-DOF Motion and High-force-torque Capability" (Co-researcher)

- Designed a compact, single-body 3-DOF (x, y, θ_z) Hybrid Reluctance Actuator (HRA) for long-stroke and high-force-torque motion.
- Implemented a flexure mechanism to ensure in-plane stability and out-of-plane constraint without compromising volumetric efficiency.
- Demonstrated rapid, full-range motion control and superior maximum force, achieving a significantly more compact form factor than conventional modular HRAs.

3. "Vibration-assisted Rapid Resin Leveling for High-speed & Recoater-free Top-down DLP" (Co-researcher)

- Developed a recoater-free Top-down DLP printing process utilizing a vibration-assisted motion strategy for rapid resin leveling.
- Simulated resin fluid dynamics during the vertical motion of the platform using MATLAB.

Undergraduate Research Intern *(Advisor: Prof. Jun Young Yoon)* Mar. 2023 – Feb. 2025
Yonsei University Seoul, Republic of Korea

1. "Modeling Force–current Formula Including Coupling Term Between DQ-axis Current and Analysis of Force Ripple in Iron-core Permanent Magnet Linear Synchronous Motor" (Personal researcher)

- Derived a PMLSM force-current model integrating d-q axis coupling terms induced by end and slotting effects.
- Analyzed periodic force ripples by applying the Magnetic Field Decomposition Method (MFDM) to the Maxwell stress tensor.
- Validated theoretical ripple characteristics via FEMM simulations, establishing a basis for ripple-reducing compensation currents.

2. "Energy-Efficient PM-Flux-Based Sensorless Soft Landing Control for Binary Electromagnetic Actuator (EMA)" (Co-researcher)

- Investigated PM-flux-based sensorless soft-landing control for binary EMAs to establish a foundation in precision motion.
- Analyzed back-EMF online position estimation and hybrid control schemes to minimize contact bounce and actuation time.
- Applied energy-efficient current profiling from this baseline to formulate advanced actuator control strategies.

3. "Foundations in Mechatronics System Modeling and Control" (Co-researcher)

- Investigated the FlexLab electromechanical platforms to establish a core foundation in the modeling, dynamics, and control of mechatronic systems.
- Analyzed multi-degree-of-freedom (DOF) motion control mechanisms utilizing on-board spiral coils as Lorentz actuators and Hall effect position sensing.

PUBLICATIONS

In Preparation

[1] **D. H. Park**, S. W. Jung, and J. Y. Yoon, "AlNiCo-based Active Stiffness Tunable XY Stage with Resonance-matched High-gain Control for Ultra-low power Large-area Scanning Motion", *In preparation*

[2] B. W. Jeon, **D. H. Park**, J. H. Park, G. H. Lee, E. K. Kim, and J. Y. Yoon, "Design and Control of Monolithic Hybrid Reluctance Actuator for Long-stroke In-plane 3-DOF Motion and High-force-torque Capability", *In preparation*

PRESENTATIONS

International Conference

[1] **D. H. Park**, S. W. Jung, and J. Y. Yoon, "Design of An AlNiCo-based Active Stiffness Tunable XY Stage with Low Steady-state Power for Resonance Match", *41st Annual Meeting of the American Society for Precision Engineering, ASPE 2026. American Society for Precision Engineering, ASPE, 2026, Submitted.*

[2] **D. H. Park**, S. W. Jung, and J. Y. Yoon, "Design and Analysis of Stiffness-tunable 2-DOF XY Stage with Enhanced Force Density", *Poster presented at the International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025).*

Domestic Conference

[1] **D. H. Park**, S. W. Jung, and J. Y. Yoon, "Ultra-Low-Power Active Stiffness-Tunable XY Stage Based on Tunable Magnet for High-Speed, Large-Area Scanning Probe Lithography and Microscopy", *Poster presented at the Proceedings of the 2026 Spring Conference of the Korean Society for Precision Engineering (KSPE).*

FELLOWSHIPS, AWARDS, AND HONORS

Fellowships

1. **NRF Master Research Grant (Student PI)**, National Research Foundation (NRF) Sep. 2025 – Aug. 2026
    - Proposal title: "Design and Control of Electromagnetic-actuator-based Active Stiffness-tunable 2-DOF Precision Stage for Ultra-low-power High-speed Scanning Systems"
    - Selected as a creative and challenging research project by the NRF of Korea.
    - Qualified as a research Principal Investigator (PI).
    - 8K USD for a year.
2. **Mechanical engineering Graduate Fellowship (MGF)**, Brain Korea 21 Apr. 2025
    - Selected as the outstanding incoming graduate student in the Department of Mechanical Engineering.
    - 5K USD.
3. **Daesang Foundation Fellowship**, Daesang Foundation Mar. 2025 – Feb. 2027
    - Selected as a talent who will contribute to national and social development.
    - 20K USD for 2 years.
4. **Academic Excellence Fellowship**, Yonsei University Spring & Fall 2022, Spring 2023
5. **Academic Excellence Fellowship**, Kyunghee University Spring & Fall 2018, Spring 2021
6. **Samsung Dreamclass Fellowship**, Samsung Welfare Foundation Dec. 2019 – Jan. 2020
    - Selected as a student to provide educational support to students from underprivileged backgrounds.

Awards

1. **Academic Excellence Award**, Yonsei University Fall 2022, Spring 2023
2. **Academic Excellence Award**, Kyunghee University Spring 2018

Honors

1. **Graduation with Honors (Top 2% of Department)**, Yonsei University
    - Selected as a representative of the Department of Mechanical Engineering at the 2025-1 Yonsei University College of Engineering degree awarding ceremony.

TEACHING EXPERIENCES

Voice Coil Motor Tutorial Tutor, Research of Manufacturing & Mechatronics Lab Mar. 2025 – Mar. 2026

- Developed tutorials on tuned mass damper (TMD) design and input shaping for a voice coil motor (VCM) stage.
- Mentored undergraduate students in classical and nonlinear control using a VCM stage, including lead–lag compensator, plant feedforward control, and GMS-based friction compensator.
- Provided hands-on instruction in mechatronics system integration, including hardware, software, and power electronics.

Teaching Assistant, Yonsei University Spring 2026

- Course: <TECHNICAL COMMUNICATION: WRITING *(Prof. Rose Richard)*>
- Corrected students` English writing.

SKILLS

Programming & Software: NI LabVIEW, MATLAB/Simulink, C, C++, Python, ANSYS Electronics, FEMM, ANSYS Workbench, Altium Designer, Autodesk Inventor, Fusion 360

Tools: Oscilloscope, Current Probe, Gaussmeter, Load cell, LCR meter, Arduino, Raspberry Pi

Language: Korean (Native), English (Fluent)